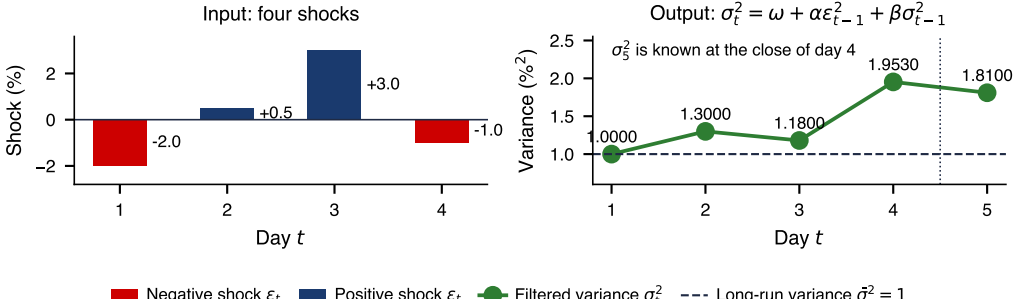
Input: four shocks
Shock (%)
2
0
−2
-2.0
+0.5
+3.0
-1.0
1
2
3
4
Day t
Output: $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$
σ_5^2 is known at the close of day 4
Variance (%^{2})
2.5
2.0
1.5
1.0
1.0000
1.3000
1.1800
1.9530
1.8100
1
2
3
4
5
Day t
Negative shock ε_t
Positive shock ε_t
Filtered variance σ_t^2
Long-run variance $\bar{\sigma}^2 = 1$